LIB

Reg. No. | | | | | | | | | | | [17EC41]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

INFORMATION THEORY AND CODING

Time: 3 Hours **Maximum Marks:70**

Instructions:

i. **Answer any FIVE full questions choosing at least TWO questions from each Part.**

ii. **Any missing data may be suitably assumed.**

PART - A

1. a. With neat sketch, explain the block diagram of an information system. **04**

b. A code is composed of dots and dashes. Assuming a dash is 3 times as long as a dot and has one-third the probability of occurrence. **06**

Calculate:

i. The information in a dot and a dash.

ii. The entropy of dot-dash code.

iii. The average rate of information if a dot lasts for 10 m-sec and this time is allowed between symbols.

c. Define entropy. State various properties of the entropy. **04**

2. a. Consider the four codes listed below. Identify the instantaneous codes and construct their individual decision trees. **07**

Source symbols	Code-k	Code-L	Code-M	Code-N
S1	0	0	0	00
S2	10	01	01	01
S3	110	001	011	10
S4	1110	0010	110	110
S5	1111	0011	111	111

b. Given the messages X_1, X_2, X_3, X_4, X_5 and X_6 with respective probabilities 0.4, 0.2, 0.2, 0.1, 0.07 and 0.03, construct a binary code by applying Shannon-fano encoding procedure. Determine code efficiency and redundancy of the code. **07**

3. a. Alphanumeric data are entered into a computer from a remote terminal through a voice-grade telephone channel. The channel has a bandwidth of 3.4Khz and output signal to noise ratio of 20 db. The terminal has a total of 128 symbols. Assume that the symbols are equiprobable and the successive transmissions are statistically independent. **07**

i. Calculate channel capacity.

ii. Find the average information content per character.

iii. Calculate the maximum symbol rate for which error free transmission over the channel is possible.

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b. State Shannon-Hartley law and show C=B log (1+S/N) bits/sec. **07**

4. a. In a communication system, a transmitter has 3 input symbols A= {a1, a2, a3} and receiver also has 3 output symbols B= {b1, b2, b3}.The matrix given below shows JPM with some marginal probabilities. **07**

$$\begin{array}{c|ccc} b_j/a_i & b_1 & b_2 & b_3 \\ \hline a_1 & 1/12 & * & 5/36 \\ a_2 & 5/36 & 1/9 & 5/36 \\ a_3 & * & 1/6 & * \\ \hline P(b_i) & 1/3 & 14/36 & * \end{array}$$

i. Find the missing probabilities (*) in the table.

ii. Find P (b3/a1) and P (a1/b3).

b. A transmitter transmits five symbols with probabilities 0.2,0.3,0.2,0.1 and 0.2. Given the channel matrix P(B/A). Calculate **i.** H(B) **ii.** H(A,B). **07**

$$P\left[B/A\right]\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1/4 & 3/4 & 0 & 0 \\ 0 & 1/3 & 2/3 & 0 \\ 0 & 0 & 1/3 & 2/3 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

PART - B

5. a. Distinguish between **06**

i. Block codes and convolution codes

ii. Random error and Burst error

b. For a systematic (6,3) linear block code, the parity matrix is $P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ Find all possible code vectors. **08**

6. a. Design an encoder for the (7, 4) binary cyclic code generated by $g(X) = 1+X+X^3$ and verify its operation using the message vectors (1001). **07**

b. Consider a (15, 11) cyclic code generated by $g(X) = 1+x+x^4$.Devise a feedback shift register encoder circuit. Illustrate the encoding procedure with the message vector 10010110111 by listing the state of the registers. **07**

7. Consider the (3,1,2) convolutional code with $g^{(1)}=(110)$, $g^{(2)}=(101)$ and $g^{(3)}=(111)$. **14**

i. Draw the encoder block diagram

ii. Find the generator matrix

iii. Find the code-word corresponding to the information sequence (11101) using time domain and transform domain approach.

8. Write an algorithm for the following coding techniques.

i. Huffman coding **07**

ii. Cyclic coding and decoding **07**